

Focused Curation Report — GLT-1 (NEUCR, Q1K4S3): "glucose transport and proton coupling"

Gene: glt-1 / NCU01633 / Q1K4S3 (*Neurospora crassa*) **Focus type:** function_assignment
Hypothesis **slug:** glucose-transport-and-proton-coupling **Disputed** **function:**
carbohydrate:proton symporter activity (GO:0005351) + proton transmembrane transport, in
addition to established low-affinity glucose uptake.

Executive Judgment

Verdict: Weakly supported / over-annotated as a specific mechanistic claim.

The seed hypothesis — that GLT-1 *retains* carbohydrate:proton symport in addition to low-affinity glucose uptake — rests entirely on **ancestry (PANTHER IBD/IBA) inference, not on any protein-specific measurement**. Three independent lines of evidence argue that the proton-symporter molecular function is not the gene product's actual activity:

1. **Annotation provenance.** Every GO term on Q1K4S3 is evidence code **IBA:GO_Central** (PANTHER PAINT), including GO:0005351. There is no experimental (EXP/IDA/IMP) support for proton coupling.
2. **Family logic.** GLT-1 belongs to the MFS **Sugar Porter family** (PTHR48022; Pfam PF00083), which contains *both* proton symporters (XylE, plant STP) *and* facilitated-diffusion uniporters (mammalian GLUT/SLC2, yeast HXT). Mechanism is **not conserved** across this family, so a family-level "sugar:proton symporter activity" cannot be assumed for a given member
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3. **Direct sequence discriminator, with an intra-organism control.** GLT-1 **lacks the acidic TM1 protonation-site residue** that is obligatory for sugar:H⁺ symport. At the position that is Asp27 in the symporter XylE and Asp40 in the symporter STP1 (and Asn29 in the uniporter GLUT1), GLT-1 has a **non-protonatable Val48** — even less symport-compatible

than GLUT. Critically, the **same scan run on the *Neurospora* high-affinity paralogs shows they RETAIN the acidic residue** — HGT-1 (Q7S0I5) Asp52, HGT-2 (Q7S3Q4) Asp35 — while only the low-affinity GLT-1 has lost it. Within one genome, proton-coupling capacity thus segregates to the high-affinity (system-II-like) transporters, exactly matching the classical physiology, and pinpoints GLT-1 as the member that specifically lost the coupling determinant. A valine cannot act as proton donor/acceptor (diagnostic per

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The seed's fair caveat holds: the identification paper

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infers system-I assignment from affinity/expression and did **not** run a GLT-1-specific proton-flux/electrophysiology assay, so it does not *experimentally* refute symport at the protein level. Hence "refuted" is too strong; but the convergent structural/evolutionary evidence makes proton-coupled symport the **less likely** mechanism. **Curation recommendation: do not assert GO:0005351 as a core/experimental function; treat it as IBA-only and preferably generalize the MF to glucose/carbohydrate transmembrane transporter activity (uniporter).** This is a lead requiring curator verification.

Evidence Matrix

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitation
UniProt Q1K4S3 (DB record, fetched 2026-09-21)	review/ database	qualifies	Basis of GO:0005351 on GLT-1	GO:0005351, GO:0008643, GO:0016020 all IBA:GO_Central ; PANTHER PTHR48022:SF92 "low affinity glucose transporter MstE"	Sugar Porter family record	High for provenance ≠/experim
P 28617850 (Ke et al. 2017)	structural/ evolutionary + biochem	refutes	TM1 residue determines symport vs uniport	"acidic residue (Asp27 in XylE) vs neutral (Asn29 in GLUT1)...key element for mechanistic divergence"	XylE/GLUT1 MD + mutagenesis	High for diagnostic multiple fine-tune
P 24847886 (Deng et al. 2014)	structural	qualifies	Same fold hosts both mechanisms	GLUT1 uniporter vs XylE proton symporter share the sugar-porter fold	Human GLUT1 crystal structure	High; est non-cons of mecha
This report — TM1 analysis (provenance CSV/PY)	computational	refutes	Does GLT-1 carry the protonation-site Asp?	GLT-1 has Val48 at the XylE-Asp27 position (XylE=D, STP1=D, GLUT1/4=N)	UniProt seqs, [FY]GY anchor	Medium single-re diagnostic wet-lab
This report — paralog control (provenance CSV)	computational/ evolutionary	refutes (for GLT-1) / competing	Does coupling segregate to high-affinity paralogs?	<i>N. crassa</i> HGT-1 (Q7S0I5)= Asp52 , HGT-2 (Q7S3Q4)= Asp35 RETAIN the acidic residue; only low-affinity GLT-1 lost it	Intra-genome paralog comparison	Medium lineage-controlled residue-I electroph

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitation
This report — expanded 11-protein panel (provenance CSV)	computational	qualifies	Is the Asp/neutral dichotomy systematic?	Symporters (XylE, STP1, HGT-1/-2)=acidic; uniporters GLUT1/3/5=neutral; GLUT2 is an acidic-but-uniporter false-positive ; GLT-1=Val (non-protonatable)	Cross-family control set	Medium; shows di is asymm absence protonat residue i robust si
P 28115989 (Wang et al. 2017)	direct assay (heterologous) + mutant	qualifies/competing	GLT-1 function & system mapping	GLT-1 = low-affinity glucose transporter (system I); HGT-1/-2 = high-affinity; radioglucose kinetics in yeast EBY.VW4000	<i>N. crassa</i> + yeast expression	High for affinity g uptake; p coupling directly a
P 24581151 (Li et al. 2014)	direct assay (heterologous)	supports (glucose uptake)	GLT-1 is a glucose transporter	NCU01633 characterized as glucose transporter in EBY.VW4000	<i>N. crassa</i>	High for specificity
P 33509260 (Li et al. 2021)	mutant/regulatory	supports (low-affinity role)	GLT-1 identity	Reaffirms GLT-1 = low-affinity transporter of the dual-affinity system	<i>N. crassa</i>	High
P 5573732 (Schneider & Wiley 1971)	direct assay (whole cell)	qualifies/competing	System I vs II energetics	System I $K_m \approx 25$ mM (low-aff, constitutive); system II $K_m \approx 0.04$ mM; only system II	<i>N. crassa</i> mycelia	High; wh (not prot specific); systems by DNP/a

Citation	Evidence type	Stance	Claim tested	Key finding	Context	Confidence/limitation
				concentrates 3-O-MG		(indirectly dependent)
P 6444943 (Rand & Tatum 1980)	direct assay (whole cell)	competing	Which system is energy-coupled	Energy-requiring, concentrative general monosaccharide transport = high-affinity system II , not system I	<i>N. crassa</i>	High for active transport in system

GO Curation Implications

- **GO:0005351 (MF, carbohydrate:proton symporter activity) — recommend: do NOT retain as an asserted/core function.** It is IBA-only, family-level, and is contradicted by the missing TM1 protonation-site residue. Options for the curator, in order of preference:
- **Generalize the MF** to an experimentally-safe, mechanism-agnostic term supported by the primary data — e.g., **D-glucose transmembrane transporter activity (GO:0055056) / glucose:proton symporter activity? NO** → prefer the non-coupled **glucose transmembrane transporter activity (GO:0005355)**. These capture the demonstrated low-affinity glucose uptake without the unsupported proton-coupling claim.
- If house style requires keeping the PANTHER IBA row, **retain GO:0005351 strictly as IBA** (do not promote to experimental, do not treat as core), and add the more precise experimentally-based glucose transporter MF above it.
- **Proton transmembrane transport (BP, GO:1902600 / GO:0015992) — recommend: remove / do not add.** No evidence; depends on the symport assumption that the sequence contradicts.
- **BP: carbohydrate/glucose transmembrane transport (GO:0008643 / GO:1904659) — retain** (well supported experimentally).
- **CC: membrane / plasma membrane (GO:0016020 / GO:0005886) — retain** (MFS integral membrane; plasma-membrane glucose uptake demonstrated).

Note on the seed's PANTHER argument: it is **factually correct** that PTN002362095 descends from the PTN000627328 IBD for GO:0005351 with no lineage-loss row, so the IBA annotation is *procedurally valid*. But procedural ancestry validity $\neq$ demonstrated mechanism; the curator should weight the protein-specific sequence and physiology evidence over an un-lost family default.

Mechanistic Scope

- **Immediate molecular function (what is actually tested):** transmembrane movement of D-glucose across the plasma membrane, low affinity (system I). Established by heterologous radioglucose kinetics

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- **The disputed add-on (coupling mechanism):** whether that glucose movement is **coupled to inward H⁺ flux** (symport, concentrative, energized by the PM H⁺-ATPase gradient) versus **facilitated diffusion** (uniport, down-gradient). This is the crux — glucose *specificity* is not in question; *coupling* is.
- **Downstream / not the tested function:** GLT-1's roles in carbon catabolite signaling, cellulase regulation, and glucose-gradient sensing

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 are pleiotropic pathway consequences, not the transport mechanism, and should not be used to justify a symporter MF.

Conflicts and Alternatives

- **Uniporter vs symporter (primary conflict).** Sequence (Val48 at the protonation site) and Sugar-Porter-family logic favor a **facilitated-diffusion uniporter**, aligning GLT-1 with low-affinity yeast HXT and mammalian GLUT rather than with H⁺-symporters. This competes directly with the IBA symporter default.
- **Whole-cell energy dependence artifact.** Schneider & Wiley (1971) found system I is *also* inhibited by azide/DNP. This could be misread as proton coupling, but uncouplers collapse

membrane potential and cellular ATP and perturb *all* PM transport indirectly; it is not evidence of stoichiometric H⁺:glucose symport by GLT-1. Only system II concentrates 3-O-methylglucose.

- **System-to-gene mapping caveat.** The historical "system I/II" physiology was measured on whole mycelia (mixtures of transporters). Equating GLT-1 exactly with all of "system I" is itself an inference; the modern gene-level data

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 support low-affinity glucose uptake but did not isolate coupling.

- **Paralog / subfamily confusion (now with positive evidence).** PANTHER subfamily label "low affinity glucose transporter MstE" (*Aspergillus*) reinforces *low-affinity glucose uniport*, not symport. The high-affinity, proton-coupling-capable activity in *N. crassa* segregates to the **HGT-1/HGT-2** paralogs, which **retain the TM1 acidic residue (Asp52 / Asp35)** that GLT-1 has lost (Val48). This is a lineage-controlled contrast within one genome: the family-default symporter GO is appropriate (at the sequence level) for the high-affinity paralogs but not for GLT-1. Care is needed not to carry the paralogs' coupling determinant onto GLT-1 via family-level IBA.

Knowledge Gaps

1. **No direct GLT-1 energetics assay.** Checked: no EXP GO, no electrophysiology in the retrieved literature. Matters because it is the only way to *prove* uniport vs symport. Resolve with: proteoliposome glucose flux $\pm \Delta\text{pH}/\Delta\psi$, or two-electrode voltage clamp in *Xenopus* oocytes measuring glucose-evoked H⁺ current.
2. **Single-residue diagnostic is asymmetric.** Checked: an expanded 11-protein panel. The signal is directional: a **non-protonatable residue (GLT-1 Val48) is strong evidence against symport** (a proton donor/acceptor is physically required), whereas an **acidic residue is necessary but not sufficient** — human GLUT2 (P11168) is a bona fide uniporter yet carries Asp27 at this position (a false-positive for "symport"). GLT-1 sits on the robust side of this asymmetry (it lacks even the neutral amide of GLUT1/3/5).

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 further cautions that multiple residues tune thermodynamics. Resolve with: full structure-based alignment / AlphaFold model + inspection of the complete proton-relay network (TM1/TM4/TM7/TM10) against Xyle and GLUT1.

3. **Membrane-potential dependence of GLT-1 specifically.** Historical energy dependence is whole-cell. Resolve with: GLT-1-only heterologous system + protonophore controls.

Discriminating Tests

- **Proteoliposome reconstitution** of purified GLT-1: measure glucose accumulation against a gradient with vs without an imposed ΔpH (inside-alkaline). Concentrative uptake requiring $\Delta\text{pH} \Rightarrow$ symport; equilibrative only $\Rightarrow$ uniport. *Most decisive.*
- **Oocyte electrophysiology:** glucose-induced inward current that is H^+ -dependent $\Rightarrow$ symport; electroneutral/no current $\Rightarrow$ uniport.
- **Structure-guided mutagenesis:** if a symport-like acidic residue can be introduced at the Val48 position and confers H^+ coupling, that would delimit the native mechanism.
- **Comparative bioinformatics (fast):** extend the TM1 protonation-site scan across all *N. crassa* Sugar Porter members and characterized fungal symporters/uniporters to confirm GLT-1 clusters with uniporters (partially done here for GLT-1 vs Xyle/STP1/GLUT).

Curation Leads (require curator verification)

- **Action change:** Downgrade/«do not assert» **GO:0005351** on GLT-1 to IBA-only or replace with mechanism-agnostic **glucose transmembrane transporter activity (GO:0005355)** / **D-glucose transmembrane transporter activity (GO:0055056)**; **do not add proton transmembrane transport BP.**
- **Retain:** carbohydrate/glucose transmembrane transport BP (GO:0008643 / GO:1904659) and membrane CC (GO:0016020), with experimental support from

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- **Candidate references + snippets to verify:**

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— "the variation between an acidic residue (Asp27 in Xyle) and a neutral one (Asn29 in GLUT1)...key element for their mechanistic divergence."

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— "Structural comparison of the uniporter GLUT1 with its bacterial homologue XylE, a proton-coupled xylose symporter..."

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— "The apparent K(m) for glucose uptake by system I and system II are 25 and 0.04 mM, respectively."

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— "a second galactose transport activity...required energy, had a high affinity...was shown to be the same as glucose transport system II."

- **Suggested curator question:** Is there *any* GLT-1-specific proton-flux/voltage-clamp datum? If not, the symporter MF should not be a core assertion.
- **Suggested experiment:** proteoliposome Δ pH-dependence assay (see Discriminating Tests).

Provenance artifacts: - `provenance/glt1_TM1_protonation_site_analysis.py` (executable) + `...analysis.csv` — core symporter/uniporter TM1 scan incl. *Neurospora* HGT-1/-2. - `provenance/glt1_TM1_expanded_panel.csv` — 11-protein control panel (documents the GLUT2 acidic-but-uniporter caveat and the diagnostic's asymmetry). - `provenance/glt1_GO_decision_table.csv` — per-term recommended curation actions.

GO decision table (summary; leads requiring curator verification):

GO term	ID	Aspect	Current	Recommended action
carbohydrate:proton symporter activity	GO:0005351	MF	IBA only	Do not assert as core; keep IBA-only or remove/generalize
D-glucose transmembrane transporter activity	GO:0055056	MF	absent	ADD (EXP-supported, mechanism-agnostic)
glucose transmembrane transporter activity	GO:0005355	MF	absent	ADD/alternative
proton transmembrane transport	GO:1902600	BP	absent	Do NOT add
carbohydrate transport	GO:0008643	BP	IBA	Retain
glucose transmembrane transport	GO:1904659	BP	absent	Add/retain
membrane	GO:0016020	CC	IBA	Retain
plasma membrane	GO:0005886	CC	absent	ADD (lead)

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